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Wind flow simulation and characteristics prediction using WAsP software for energy planning over the region of Hassi R'mel

Fares Meziane [a,b], Farouk Chellali [c], Kamal Mohammedi [a], Ilyes Nouicer [b], and Nourdine Kabouche [b]

[a]URMPE/MESO, University M'Hamed Bougara, Boumerdès, Algeria; [b]Centre de Développement des Energies Renouvelables, Algierss, Algeria; [c]Department of Electrical Engineering, University Ziane Achour, Djelfa, Algeria

ABSTRACT

In this study, we analyzed the wind potential of Hassi R'mel (Algeria) under the Wind Atlas Analysis and Application Program (WAsP) environment, using data measurements at a height of 10 m provided by the National Office of Meteorology (NOM) over the period of 14 years (2003–2017). The results are used as input for predicting wind characteristics (speed and direction) for four other sites covering the studied region where the measurement stations are not available. The pre-feasibility study of large-scale energy and hydrogen production focuses on four sites, namely Telghemt, Delaa, Bouzbier, and Bellil. Two different wind turbine models, Bonus B54 and Nordex N100 with a rated power of 1000 kW and 2500 kW, respectively, were selected. The study pointed out that the region of Hassi R'mel has an important wind potential with an annual mean wind speed of 6.73 m/sec and power density of 365 W/m^2 at 10 m. The maximum of energy and hydrogen produced occurs for the site of Hassi R'mel with 12.41 GWh/yr, and 177.37 tons H_2/yr, with minimal costs of 5.5 c$/kWh, and 18.42 $/kg H_2 respectively using Nordex N100 wind turbine at 100 m. Similarly for the site of Bellil where the maximum of energy and hydrogen production of 5.395 GWh/yr, and 77.08 tons H_2/yr were obtained with minimal costs of 5.06 c$/kWh, and 17.12 $/kg H_2 respectively using Bonus B54 wind turbine at 60 m. Therefore, the high rate of carbon dioxide (CO_2) emissions avoided by N100 was 5428.2 tons CO_2/kWh for Hassi R'mel, and 2359.2 tons CO_2/kWh for Bellil using the B54 wind turbine.

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CO_2; cost; energy; Hassi R'mel; hydrogen; potential; predicting; WAsP

1. Introduction

For decades, the Algerian energy mix relied on fossil energy sources, where most electricity production is generated from natural gas combustion despite the important potential of renewable energy, particularly wind energy (Boudghene Stambouli et al. 2012), which can provide clean energy while minimizing the environmental impacts. The declining oil and gas resources and the commitments to low carbon mid-century strategies, makes Algeria facing a challenging choice ending its dependence on fossil fuels and its capacity to shift to renewable sources. Promoting the use of wind energy requires an accurate evaluation of this resource, and a better understanding of its distribution. In this context, few studies on wind energy potential assessment concerning Algeria have been carried out (Said and Ibrahim 1984), and (Bensaid 1985). The first wind map of Algeria was proposed by Kasbadji Merzouk (2000). Then, wind speed distributions map at 10 m above the ground level was established by (Aîche-Hamane and Khellaf 2003) using monthly averaged wind speeds. Since then, an improved wind map of Algeria (Kasbadji Merzouk 2006) was established by increasing the number of measurement sites. Chellali et al. (2011) updated the wind map of Algeria by adding data for the region of Hassi R'mel; followed by the contribution of (Boudia et al. 2013) using more recent average daily speeds database with larger weather stations. Recently, Nedjari et al. (2018) discussed optimal windy sites in Algeria, using wind data of 21 weather stations.

Modeling wind flow allows predicting important characteristics of the wind resource at locations where measurements are not available or incomplete. Physical and statistical-based methods dealing with wind prediction have been used in many studies. In physical methods, Computational Fluid Dynamics (CFD) takes into consideration terrains topography and roughness, and utilizes the output from Numerical Weather Prediction (NWP) models which provide weather forecasts. These models based on Navier-Stokes equations resolution take into consideration nonlinear effects and the flow dividers (Montavon 1998). Furthermore, the statistical methods are based on historical meteorological-measured database, wind farm operation data, and other possible influencing factors to predict wind-resource potential. Bilgili, Sahin, and Yasar (2007) used hourly wind speed data provided by eight neighboring measuring stations to predict the wind speed for a given station by applying the artificial neural networks (ANN) method. They conclude that the ANN method is more suitable for wind speed predicting for a given station. The ANN method is more suitable for wind speed predicting for a given station. The ANN method accuracy depends on the correlation between the target and reference stations, and on the availability of the wind speed data of the reference stations. Erdem and Shi (2011) proposed four different approaches based on the autoregressive moving average (ARMA) method for short-term wind speed and direction forecasting, and assessed their performance using the metric of mean absolute error (MAE).

CONTACT Fares Meziane fares.meziane1981@gmail.com URMPE/MESO, M. Bougara university, Boumerdès 35000, Algeria.

The component model is better suited in predicting the wind direction, while the traditional-linked ARMA model is recommended for wind speed forecasting. Bilgili and Sahin (2011) compared results and analyzed the accuracy of linear regression (LR), nonlinear regression (NLR), and artificial neural network (ANN) methods on wind speed prediction. The ANN methods were found to provide better performance than LR and NLR methods. Okumus and Dinler (2016) reviewed recent advances in statistical wind forecasting, and listed different prediction methods. To improve the stability and accuracy of wind power forecasting, we can either combine different methods, or use different numerical programs. Thereby, the most common environment used for wind-resource predictions is the Wind Atlas Analysis and Application Program (WAsP) (Lange and Højstrup 2001; Mortensen et al. 1993) based primarily on the concept of linear flow models (Jackson and Hunt 1975), and considering as inputs the representative wind data collected at the meteorological stations. It uses an interpolation technique to predict wind potential at other sites. A preliminary wind Atlas for Algeria based on WAsP was done by Hammouche (1990). Linear-based models previously used for establishing wind maps have produced some reliable estimation of wind speeds over a flat terrain (Walmsley, Taylor, and Keith 1986). Some other works dealing with the use of WAsP were considered, Mezidi et al. (2019a) studied the wind power generation in Adrar and Tamanrasset. Boudia et al. (2013, 2014, and 2018); Boudia and Guerri (2015) investigated the wind power potential, and assessed the wind energy for different regions in Algeria. A significant study dealing with wind farm feasibility was carried out by (Djamai and Kasbadji Merzouk 2011). The authors discussed the feasibility of installing a 10 MW wind farm in the south-western region of Algeria characterized by important wind potential using WAsP software. This study pointed out that in terms of wind resources (wind speed reaching 9 m/sec at 80 m), accessibility, low density of population, significant-naked spaces, and absence of protected zones, the Adrar region was the most suitable location for wind farm installation. Three years later, a 10 MW wind farm project was achieved Kabertene (northern of Adrar city). In fact, this farm is considered relatively new (2014) to assess the impact of the desert climate on it, but most of the difficulties in running it lies in the penetration of sand grains in addition to the high temperatures that force them to stop most of the time. Afterward, an optimization study of the Kabertene wind turbines location was done (Benmbarek, Benzergua, and Khiat 2015) using WAsP software.

Recently, Guerri et al. (2020) compared the accuracy of predicted performances obtained using power density functions (PDFs) to actual data of Kabertene wind park. The study pointed out that, the generalized extreme value (GEV) distribution is more suitable for estimating the annual energy production (AEP), while the Weibull and Normal distributions are equivalent for the estimation of the average wind speed.

Ensuring a continuous supply of power for a given region is also a challenge, due to the intermittent nature of wind energy. Nevertheless, this inconvenience can be addressed by using an excess of energy to produce hydrogen for energy storage. Hydrogen can then be used when wind energy is not sufficient or not available. Mantz and De Battista (2008) investigated the possibility of hydrogen production using the excess of energy generated by wind turbines, and developed an adequate control strategy considering the system operating parameters to achieve the grid requirements. Korpa and Christopher (2008) have shown different possibilities of combining wind and hydrogen in low production networks. They examined the benefits, and hydrogen storage applications limits, and they presented the performance of wind-hydrogen systems. The results illustrate the benefits of storing hydrogen and its use for electricity generation during low wind speed periods. Khan and Iqbal (2009) presented a detailed modeling, simulation, and analysis of an isolated wind-hydrogen hybrid energy system. They concluded that the profitability of stand-alone energy systems, with hydrogen storage and generation is not favorable for mass deployment. However, research and advancements in this field are quite dynamic and the cost-competitiveness of wind-hydrogen systems may improve in the future. Bechrakis, McKeogh, and Gallagher (2006) simulated the operation of a small remote hotel primarily powered by a wind turbine, and supported by a hydrogen energy system. The simulation results showed that this particular system would meet the electrical energy needs of a 10 bedroom hostel. Olateju and Kumar (2011) made an assessment of hydrogen production from wind energy in Western Canada by combining a 1.8 MW wind turbine and electrolyzer with 240 KW, and 360 KW nominal power. The minimum hydrogen delivery cost has been found to be 4.96 \$/kg H_2. The life cycle of CO_2 emissions is 6.35 kg CO_2/kg H_2. Rodriguez et al. (2010) investigated hydrogen production potential from wind energy and the use of the produced hydrogen as a fuel in the transport sector. The annual energy demand of the transport sector was completely covered by the hydrogen production. Genc, Celik, and Genc (2012) considered the case of hydrogen production in Pınarbası Kayseri using a wind-electrolysis system. They estimated the potential and cost of hydrogen production for different hub heights, using electrolyzers with different-rated power values. The maximum hydrogen production was 14192 kg H_2/year; while the minimum hydrogen cost of 8.5 \$/kg H_2 was obtained for a wind energy conversion system with 100 m hub height. Genc, Celik, and Karasu (2012) proposed a review for the investigation of hydrogen production from wind energy and hydrogen production costs in Turkey, for different heights at five locations in Turkey. Pinarbasi and Sinop sites have remarkable wind potential. Gökçek and Genç (2009) established a preliminary study on wind energy cost in Central Anatolian–Turkey by considering small-scale wind turbine models. It has been found that the levelised cost of electricity varies between 0.29 and 30.0 \$/kWh using a time-series approach for all cases.

In Algeria, few studies dealing with clean and sustainable hydrogen production from renewable energies have been carried out; Boudries and Dizene (2008, 2011, and 2013) discussed the different methods for hydrogen production from renewable energy sources, and estimated the Algeria's potentialities of sustainable and renewable hydrogen, and reviewed the Algerian energy situation. They also discussed the perspectives of using hydrogen in the national energy mix. Aiche-Hamane, Hamane, and Belhamel (2007, 2009) studied the possibility of hydrogen production from wind energy through water electrolysis in the south of Algeria. Recently, Douak and Settou (2015) discussed the production of electrolytic hydrogen from the wind energy source in the three most windy regions in the south of Algeria. Using different wind turbine sizes, the low cost of hydrogen is occurred for the site of Adrar with 1.214 $/kg using “De Wind D7” wind turbine.

2. Methodology

2.1. Study objective

The present study is a continuation of the previous work initiated by Chellali et al. (2010and 2011) who updated the wind map of Algeria. His study has focussed on the region of Hassi R’mel which was not previously identified as a windy region because it lies between two less windy regions (Ghardaia and Laghouat). Indeed, this region is characterized by wind speeds exceeding 6 m/sec at 10 m above ground level. This study relies also on the work previously presented by (Meziane et al. 2020) considering the estimation of the wind resource as well as the energy and hydrogen production only for the site of Hassi R’mel where the meteorological measurement station is available. The main objective of the present paper is to characterize the region of Hassi R’mel, and predict the characteristics of wind flow at different sites covering this region. The first part of the study concerns the evaluation, analysis, and wind-resource mapping at 10 m based on the measured wind data provided by the NOM at the site of Hassi R’mel from 2003 to 2017 using WAsP software; the subsequent sections are focusing on the utilization of WAsP software to predict wind speeds and directions for four sites covering the selected area, where the wind data are not available, namely, Telghemt, Delaa, Bouzbier, and Bellil. Economic and environmental analysis is proposed for each site to identify the suitable sites for large-scale energy and electrolytic hydrogen production. The different steps are summarized in Figure 1.

2.2. Selected area

The region of Hassi R’mel is located in the northern Sahara of Algeria, characterized by heavy sand storms and desert climate, the altitude is varying from 600 m to 850 m. It is bounded by longitudes ranging from 3.1° to 3.6° East and latitudes varying between 32.8° and 33.5° North. The two-dimensional map of the Hassi R’mel region was established under Matlab tool using SRTM (Shuttle Radar Topography Mission) database as shown in Figure 2.

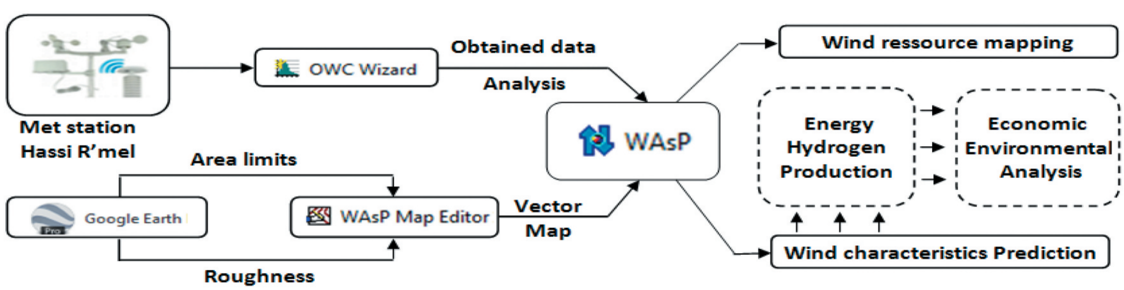

Figure 1. Characteristics prediction using WAsP for Hassi R’mel region.

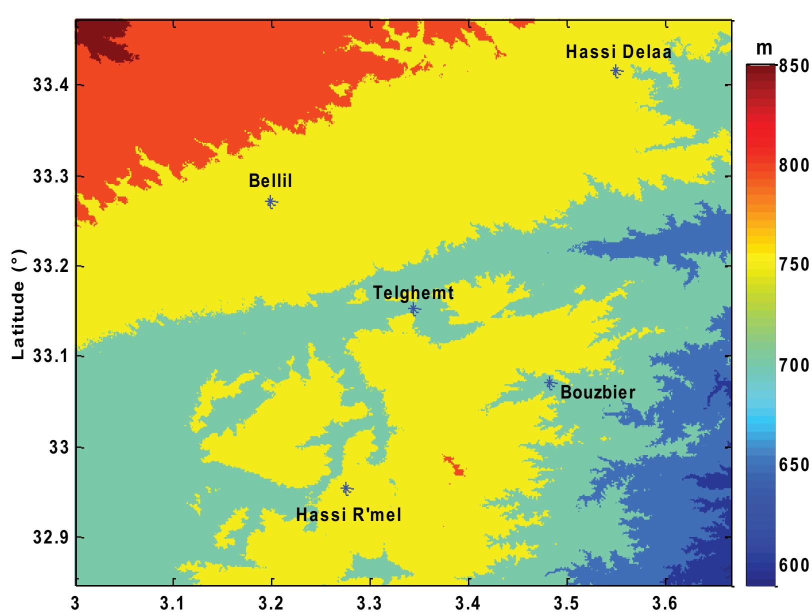

Figure 2. SRTM data base topographical map of Hassi-R’mel region.

2.3. Wind data

This study is based on hourly measurements of wind speed and direction at 10 m above ground level (a.g.l) collected over 14 years period from 2003 to 2017, at Hassi R'mel meteorological station (32.95°E, 3.92°N, 682 m).

2.4. Wind turbine characteristics

We selected two turbine models for large-scale capacity, Bonus B54 (1000 kW), and Nordex N100 (2500 kW). The chosen wind turbines start producing power at the same cut-in speed of 3 m/sec. The cutout speeds of the machines are 25 m/sec and 20 m/sec, respectively. The technical data of the chosen wind turbines are shown in Table 1.

2.5. Modeling

The Weibull distribution used to describe the wind speed distribution is defined by the following equation (Boudia and Guerri 2015):

$$f(V) = \left(\frac{k}{V}\right)\cdot\left(\frac{V}{C}\right)^{k-1} \exp\left[-\left(\frac{V}{C}\right)^{k}\right] \tag{1}$$

The scale factor c (m/sec), and the shape parameters (k) of the Weibull distribution are done as follow:

$$k = \left(\frac{\sigma}{\mathrm{Vm}}\right)^{-1.086} \tag{2}$$

$$C = \frac{Vm}{\Gamma\left(1+\frac{1}{k}\right)} \tag{3}$$

The power density as a function of the Weibull parameters can be obtained using the following formula (Ahmadian et al. 2013):

$$PD = \frac{1}{2}\times\rho\times C^3\times\Gamma\left(1+\frac{3}{k}\right) \tag{4}$$

The annual energy (AEP) represents the power produced by the wind turbine during the year, this generated energy can be converted to hydrogen via electrolysis; the amount of hydrogen mass M_{H2} (kg/yr) estimated, is given as follows (Douak and Settou 2015):

$$M_{H2} = \frac{\eta_{el}\times AEP}{LHV_{H2}} \tag{5}$$

LHV_{H2} is a low heat value of hydrogen 121 (MJ/kg).

2.6. WAsP software overview

WAsP is a software program used for wind-resource assessment, wind turbines siting, and wind farms. It also enables to predict of wind climates, wind resources, and power productions from wind turbines and wind farms. The predictions are based both on wind data measured on-site or measurement stations in the same region considering topography effects on the wind flow. It includes a complex terrain flow model, a roughness change model, and a model for sheltering obstacles.

Table 1. Technical specifications of the considered wind turbines.

Turbine	Hub_h	Rat_pow	cut_in	Rat_s	cut_off	Rot_diam
Bonus B54	60	1000	3	15	25	54.2
Nordex N100	100	2500	3	14	20	100

3. Economic analysis

The cost of a wind turbine (C_{wt}) is the product of the specific cost and the rated power (Diaf, Notton, and Diaf 2013):

$$C_{wt} = C_{spe}\times P_r \tag{6}$$

Table 2 shows the specific cost of turbines (C_{spe}) for different size ranges based on the wind turbine rated power (Pr); it consists to consider the average between the maximum and the minimum value as given in the following table.

According to this table, for wind turbines with rated power above 200 kW, the specific cost value represents the average of the minimum ($/kW 700), and the maximum ($/kW 1600) specific cost, it is assumed to be 1150 $/kW.

The cost of the electrolyzer depends on the electrolyzer unit cost (C_{elec_u}), the quantity of hydrogen mass production (M_{H2}), the electrolyzer efficiency (η_{el}), and the required electrolyzer's theoretical specific energy (K_{el_th}) as indicated by the following equation (Douak and Settou 2015):

$$C_{elec} = C_{elec\_u}\times\frac{M_{H2}\times K_{el\_th}}{8760\times\eta_{el}} \tag{7}$$

For our case, the theoretical specific energy required by the electrolyzer (K_{el_th}) and the electrolyzer unit cost (C_{elec_u}) were considered to be 39.7 kWh/kg H_2, 368 $/kW respectively.

The present value of costs (PVC) of electricity includes both initial costs (I_C), and operation, maintenance, and repair cost (C_{omr}), is given by the following formula (Ahmed Shata and Hanitsch 2006; Bagiorgas et al. 2007):

$$PVC = I_C + C_{omr}\times\left[\frac{1+i}{r-i}\right]\times\left[1-\left(\frac{1+i}{1+r}\right)^t\right] - S \times\left(\frac{1+i}{1+r}\right)^t \tag{8}$$

The I_C ($) includes the turbine price ($), civil work ($), and installation costs ($), the civil works are assumed to be 20% of the turbine price. The annual operation maintenance costs (C_{omr}) was considered to be 15% of the annual cost of the wind turbine (Diaf, Notton, and Diaf 2013). The cost of energy (COE) in ($/kWh) is the ratio of the present value of costs (PVC) of the project on produced energy:

$$COE = \frac{\mathrm{PVC}}{AEP} \tag{9}$$

Table 2. The specific cost of wind turbines based on the rated power (Diaf, Notton, and Diaf 2013).

Wind turbine size P (kW)	Specific cost ($/kW)	Average value ($/kW)
Pr<20	2200–3000	2600
20< Pr<200	1250–2300	1775
Pr>200	700–1600	1150

The cost of hydrogen (COH) in ($/kg) is calculated by combining Equations (5)–(7) during lifetime (t) as follows (Douak and Settou 2015):

$$COH = \frac{C_{elec} + C_{wt}}{M_{H2} \times t} \tag{10}$$

4. Carbon dioxide emissions analysis

Estimating carbon dioxide emissions, is an important task to assess the impact on the environment caused by the energy generated from different sources (fossil, nuclear, and renewable), and makes better decisions referring to environmental protection. The CO_2 rate emitted by different energy sources is given in Table 3.

Mainly, this section aims to compare the CO_2 level emitted by two different wind turbines with that emitted by natural gas plants as the main source of energy for the region of Hassi R'mel, and showing the advantages of wind energy in terms of reducing the rate of CO_2 emitted. In this case, Wang et al. (2012) pointed out that the CO_2 rate emissions range was 5.0–8.2 (g CO_2/kWh) for wind turbine plants, which is significantly lower than estimates in previous studies that considering individual wind turbines with different-rated power. The Renewable Energy Report analyzing the emissions generated by wind energy, published by the Intergovernmental Panel on Climate Change (IPCC 2011), shows that one kWh of electricity generated from wind turbines would produce between 2 and 80 g of CO_2 equivalent with a median value below 25 g. A recent study established by ADEME (Cycleco 2015), pointed out that the emission rates for wind farm onshore and offshore are 12.7 (g CO_2 eq/kWh), and 14.8 (g CO_2 eq/kWh) respectively.

In the present study, according to (Haddouche and Boudia 2019) we considered that 1 kWh of energy produced from wind energy (onshore) can emit 12.7 (g CO_2/kWh), and the average rate of CO_2 emitted by natural gas plants to produce 1 kWh of energy was 450 (g CO_2/kWh) (Ardente et al. 2008; Haddouche and Boudia 2019).

5. Results and discussions

5.1. Wind potential at 10 m

Monthly average wind speeds at 10 m (anemometer measurement height) are varying from 5 m/sec to 8 m/sec as indicated in Figure 3. High wind speeds are recorded during the spring season (March, April, and May) which is characterized by sand storms, leading to reach a high wind speed of 8.5 m/sec in May. Otherwise, the lowest wind speed is recorded for July with 4.97 m/sec, with a significant shift in wind speeds exceeding 3.5 m/sec. The annual average wind speed is 6.73 m/sec.

Table 3. Environmental performances of plants (Ardente et al. 2008; Haddouche and Boudia 2019).

Electric generation plants	Global warming potential ($g_{CO2,eq}$/kWh)	Average value ($g_{CO2,eq}$/kWh)
Coal plants	900–1200	1050
Diesel, heavy oil plants	780–900	840
Natural gas plants	400–500	450
photovoltaic	50–100	75
nuclear plants	15–50	32.5
hydroelectric plants	15–40	27.5

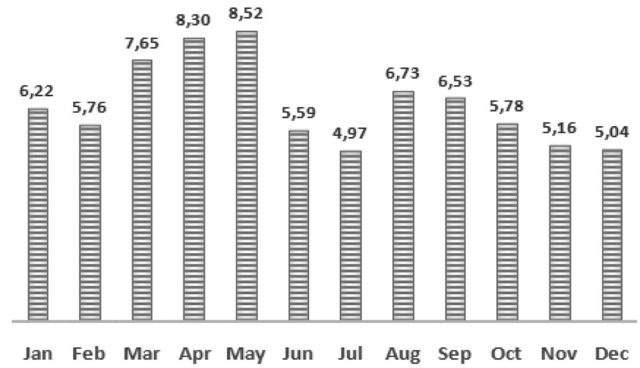

Figure 3. Monthly distribution of wind speeds at 10 m.

Regarding the annual histogram of wind speeds shown in Figure 4, it can be seen that the wind speed covers a range of variation reaching 20 m/sec, while the wind speeds between 6 and 8 m/sec are the most frequent and represent the highest rate of 16%. The high value of the scale parameters (C = 7.6 m/sec) means that this site is windy.

It is also noticed that the Weibull shape parameter k reaches 1.98 corresponding to regular and steady winds, with an annual average wind speed of 6.73 m/sec, and the annual mean power density of 365 W/m^2 that makes us predicting an important annual wind resource for this site.

The wind rose plays an important role in determining the most suitable location and more profitable direction for wind turbine system installation.

The wind rose given in (Figure 5) indicates that the dominant directions over the whole-studied period where the frequencies are highest for the site of Hassi R'mel are, the East with 7%, North, south, and South-West with 6.9%.

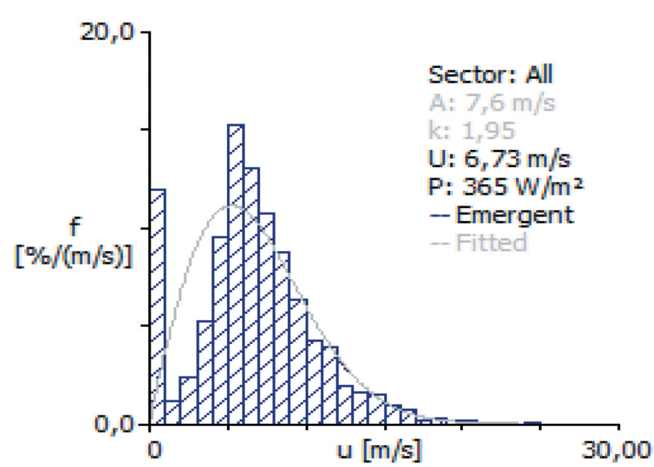

Figure 4. Weibull distribution at 10 m.

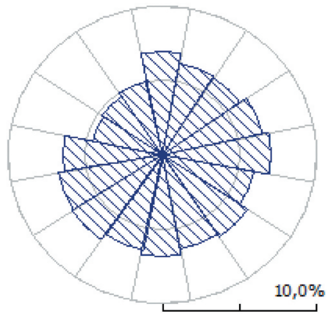

Figure 5. Wind Rose diagram at 10 m.

5.2. Wind-resource mapping at 10 m

The Wasp software is used to perform the wind-resource mapping for the region of Hassi R'mel (40 km x 60 km). The mean wind speed distributions, and power density maps at a height of 10 m (a.g.l) with a resolution of 100 m, for the region of Hassi R'mel, are established based on wind statistical data, the digital elevation map, and the roughness length map, as illustrated in (Figures 6 and 7), showing that the annual average wind speed at 10 m a.g.l varies from 6.5 m/sec for the less windy areas that corresponding to a power density of 300 W/m^2, and reaching 11.5 m/sec for the windiest locations of the region that approximately corresponding to the high power density of 2000 W/m^2.

5.3. Wind characteristics predicting

For our case study, WAsP predictions are based on the observed wind climate (measured wind speeds and directions) at Hassi R'mel weather station at 10 m over 14 years (2003–2017), that have been binned into wind rose, and wind histograms, in order to predict the wind characteristics at four other sites covering this region where the measurement stations are not available; namely, Telghemt, Delaa, Bouzbier, and Bellil, see Figure 2. The geographical coordinates of the different sites are illustrated in Table 4.

5.3.1. Wind speed and power density for considered hub heights

WAsP is designed in order to be able to assess and predict the wind resource or wind power potential, at one or several sites, or over an area, based on measured wind data to establish wind-resource map or bins showing average wind speed variation or power density, for a given height above ground level.

Figure 8, shows that at 60 m of height, the predicted wind speeds vary from 10.94 m/sec for Telghemt to 12.82 m/sec for Bellil with a difference of 1.88 m/sec. However, at 100 m the low wind speed is occurring for the site of Telghemt with 12.23 m/sec and the high wind speed of 13.69 m/sec for the site of Bellil with a difference of 1.46 m/sec.

By comparing the predicted wind speeds of other sites to Hassi R'mel as a referential site, it can be noted that, for both hub heights 60 m and 100 m, the high rate of increasing wind speed has occurred for the site of Bellil with 17% and 11%, respectively. Except for the site of Telghemt where the increasing predicted wind speed is 0 at 60 m, and decreasing by 9% at 100 m.

As illustrated in Figure 8, at the height of 60 m the estimated power density varies between 1499 W/m^2 for the site of Hassi R'mel and 2455 W/m^2 for the site of Bellil. Hence, at the height of 100 m, the minimum power density of 2091 W/m^2 is observed for Hassi R'mel site, and a maximum of 2900 W/m^2 for the site of Bellil.

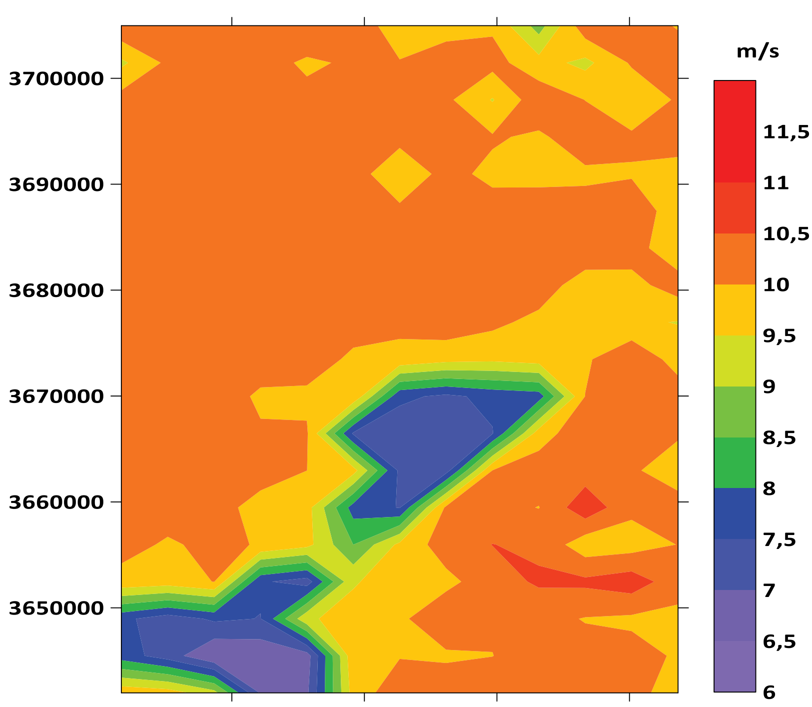

Figure 6. Wind speed evolution over the studied region at 10 m.

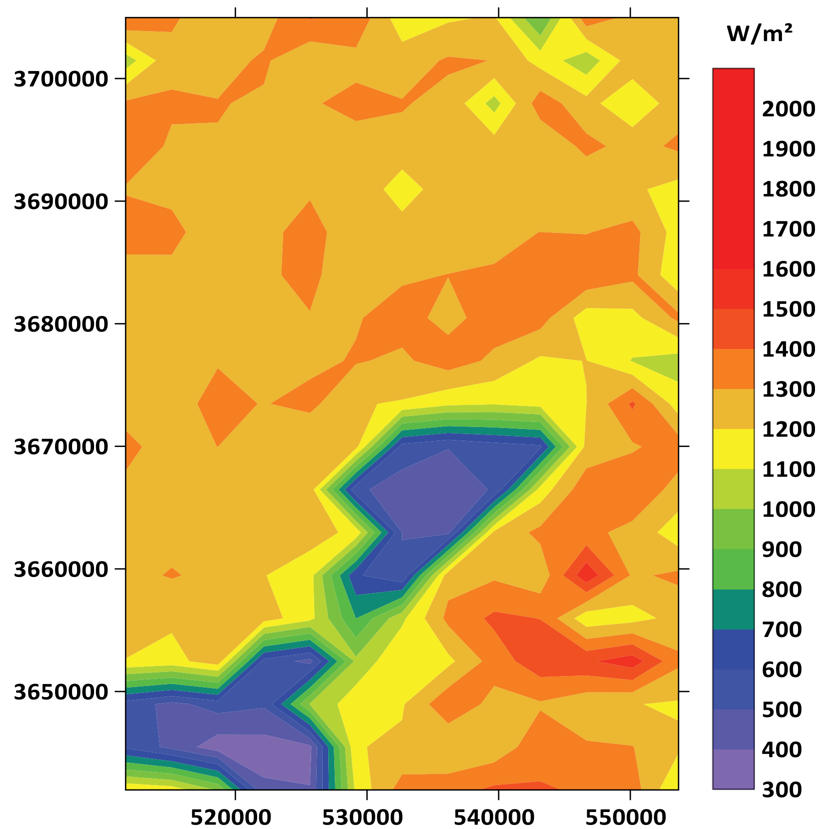

Figure 7. Power density estimated for the studied region at 10 m.

Table 4. Geographical coordinates of under-considered sites.

Location	Telghemt	Delaa	Bouzbier	Bellil
Latitude	33.26° N	33.42° N	33.07° N	33.27° N
Longitude	03.33° E	03.54° E	03.48° E	03.19° N
Altitude	758 m	757 m	735 m	778 m

Table 5. Yearly energy calculated (GWh/yr).

Site	Height	H-R'mel	Telghemt	Delaa	Bouzbier	Bellil
B54	60 m	4.96	4.85	5.34	5.36	5.40
N100	100 m	12.41	12.24	12.27	12.25	12.18

5.3.2. Energy and hydrogen production considering hub heights

For the two selected turbine models at their respective hub heights, the yearly estimated energy output is estimated in Table 5, where the energy produced is varying from the minimum of 12.18 (GWh/yr) for the site of Bellil to the maximum of 12.41 (GWh/yr) for the site of Hassi R'mel using Nordex N100 wind turbine at the hub height of 100 m. However, the minimum of energy produced is observed for the site of Telghemt with 4.848 (GWh/yr) and the maximum of

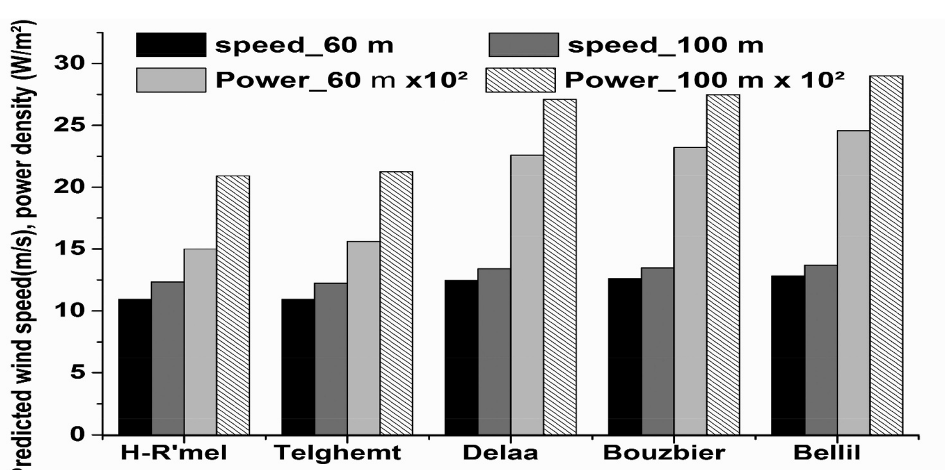

Figure 8. Predicted wind speed and power density at 60 m and 100 m.

5.395 (GWh/yr) for the site of Bellil using Bonus B54 wind turbine at the hub height of 60 m.

From Table 6, we noticed that the amount of estimated hydrogen production is varying between 64.13 (tons H_2/yr) and 77.09 (tons H_2/yr) for Bonus B54. However, for Nordex N100 the amount of hydrogen produced varies between 174.10 (tons H_2/yr) and 177.38 (tons H_2/yr).

Analyzing the results of Tables 5 and 6, we conclude that the highest production of energy and hydrogen is occurring when using Nordex N100 wind turbine at 100 m for the site of Hassi R'mel with 12.413 GWh/yr, 177.37 tons H_2/yr respectively. Hence, Bonus B54 wind turbine at 60 m produced the maximum of energy and hydrogen for the site of Bellil with 5.395 GWh/yr, 77.09 tons H_2/yr respectively.

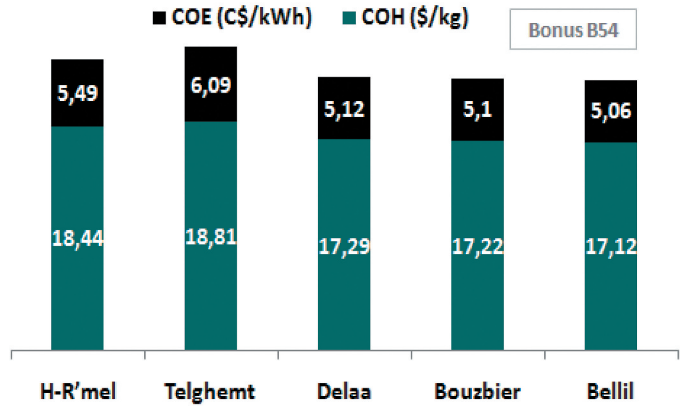

Figure 10. Energy and Hydrogen costs for Bonus B54.

5.4. Cost analysis

The costs of energy and hydrogen are calculated for the considered sites covering the region of Hassi R'mel using Bonus B54 and Nordex N100 wind turbines at 60 m, 100 m, respectively.

Considering Figures 9 and 10, the low cost of energy and hydrogen is observed for the site of Hassi R'mel, with 5.5 c $/kWh and 18.42 $/kg, respectively, using Nordex N100 wind turbine at 100 m. Besides, the low cost of energy and hydrogen is occurred for the site of Bellil with 5.06 c $/kWh and 17.12 $/kg, respectively, using Bonus B54 wind turbine at 60 m.

5.5. Carbon dioxide emissions analysis

In this section, the carbon dioxide (CO_2) rate emitted by two different wind turbine models (Bonus B54, and Nordex N100) for each site was estimated, then compared to that emitted by a gas plant considering as the main source of energy for the region of Hassi R'mel. Considering Table 3 values, the global warming potential for natural gas plants varies between 400 and 500 (g CO2 eq/kWh). In this case study, we considered the average value of CO_2 emitted of 450 (g CO_2 eq/kWh). Therefore, the avoided CO_2 emissions were considered as the difference between the global warming potential emitted by natural gas plants indicated in Table 3, and that emitted by the wind source (calculated from Table 5), and based on results given by (Cycleco 2015).

Table 6. Yearly hydrogen production estimated (tons H_2/yr).

Site	Height	H-R'mel	Telghemt	Delaa	Bouzbier	Bellil
B54	60 m	71.09	64.13	76.26	76.59	77.09
N100	100 m	177.37	174.86	175.28	174.97	174.10

In Figure 11, we notice that the avoided CO_2 emission rate using Nordex N100 wind turbine is varying from 5328.1 (tons CO_2/kWh) for the site of Bellil and 5428.2 (tons CO_2/kWh) for the site of Hassi R'mel, respectively, with the difference of 100 (tons CO_2/kWh). However, for Bonus B54 wind turbine, the high rate of avoided CO_2 emission rate is recorded for the site of Bellil with 2359.2 (tons CO_2/kWh), and the low avoided CO_2 rate of 2120 (tons CO_2/kWh) is recorded for the site of Telghemt, with a difference of 239.2 (tons CO_2/kWh). Comparing the two wind turbine models, we conclude that the use of Nordex N100 wind turbine avoids an average additional rate of 3100 (tons CO_2/kWh) compared to Bonus B54 for every

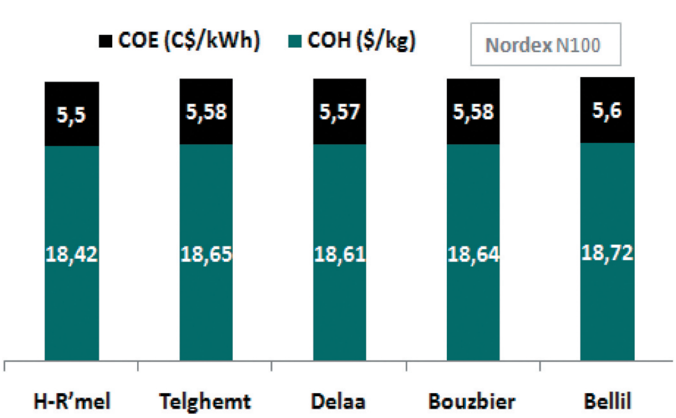

Figure 9. Energy and Hydrogen costs for N100.

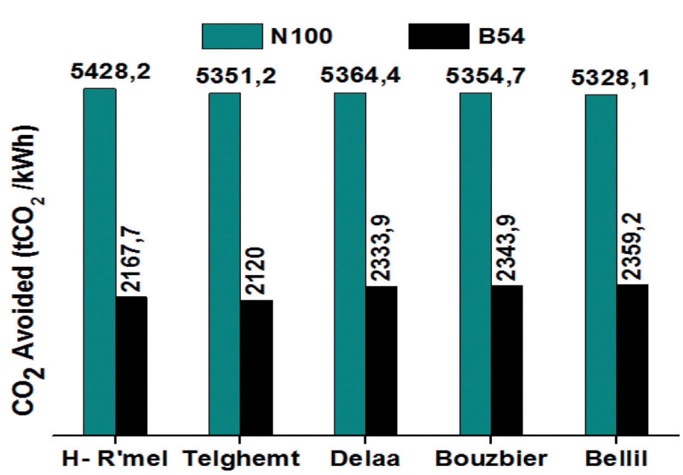

Figure 11. CO2 emission avoided comparison for N100 and B54.

site. For both wind turbines and for all sites the gain in terms of CO_2 rate emission is 97.17%.

6. Conclusion

In this study, we use the WAsP simulation environment to analyze Hassi R'mel site wind data and estimate wind potential using data provided by NOM at 10 m above ground level. The average wind speed is around 6.73 m/sec, the average power density is 365 W/m^2, the scale parameter C = 7.6 m/sec, the shape parameter is k = 1.95 obtained at 10 m according to the Weibull distribution. Hence, the dominant wind direction has been observed as North at 0° sector, where the average power density was 388 W/m^2, the Weibull parameters C and k were 7.8 m/sec, 1.97, respectively.

The analysis result is necessary inputs for the WAsP environment to predict wind characteristics for four sites covering the region of Hassi R'mel where measurement data are not available, namely, Telghemt, Delaa, Bouzbier, and Bellil. The accurate prediction of wind energy distribution is an important task for making an appropriate selection of suitable sites and facilitates decision-making for wind farms implementation. We obtained preliminary results for large-scale energy and hydrogen production using wind energy sources over this region, by considering two different wind turbine models, Bonus B54, and Nordex N100 with 1000 kW, 2500 kW rated power, respectively. The economic and environmental analyses for each site were carried out. The results pointed out that the maximum annual energy produced is 12.413 GWh/yr with a low cost of 5.5 c$/kWh for the site of Hassi R'mel using N100 at a height of 100 m, and 5.39 GWh/yr with a low cost of 5.06 c$/kWh for the site of Bellil using Bonus B54 at 60 m.

Higher production of hydrogen was observed for the site of Hassi R'mel with 177.37 tons H_2/yr for a minimal cost of 18.42 $/kg H_2 using N100 at 100 m, and 77.08 tons H_2/yr with 17.12 for the site of Bellil using B54 at 60 m. The cost analysis of energy and hydrogen has successfully demonstrated that wind energy is another alternative option to adopt in Hassi R'mel, which shows a remarkable potential of wind energy in addition to significant solar energy that can be economically profitable, it is only a matter of time to replace fossil energies.

The annual energy production and its cost are influenced by the data period measurements (Mezidi et al. 2019b), for both wind parks case study situated in Adrar and In Salah and for two different data measurement periods, the annual energy production (AEP) decreases while the low cost of energy (LCOE) increases rapidly. Moreover, the maximal values of the LCOE for Adrar are 6.42, 4.97 c$/kWh, and 3.67, 2.85 c$/kWh for In Salah. In term of costs, the comparison proves difficult because of its dependence on the measurement period, for our case study, the obtained results are acceptable.

Nomenclature

ADEME	Agence de l'environnement et de la maitrise de l'énergie
AEP	Annual Energy Production (kWh/yr)
a.g.l	Above ground level
ANN	Artificial Neural Networks
ARMA	Autoregressive Moving Average
C	Weibull scale parameter (m/sec)
C_{elec}	Electrolyzer cost ($)
C_{elec_u}	Electrolyzer unit cost ($/kWe)
CFD	Computational Fluid Dynamics
CO_2	Carbon dioxide
COE	Cost of energy ($/kWh)
COH	Cost of hydrogen ($/kg)
C_{omr}	Maintenance and repair cost ($)
C_{spe}	Specific cost of turbines ($/kW)
Cut_in	Cut in speed (m/sec)
Cut_off	Cut off speed (m/sec)
Cwt	Cost of wind turbine ($)
Hub_h	Hub height (m)
i	Inflation rate (%)
Ic	Initial cost ($)
IPCC	Intergovernmental Panel on Climate Change
k	Weibull shape parameter
Kel_th	Electrolyzer's theoretical specific energy (kWh/kg)
LHV_{H2}	Low Heat Value of Hydrogen (MJ/kg).
LR	Linear Regression
MAE	Mean Absolute Error (%)
M_{H2}	Hydrogen mass (kg/yr)
NLR	Non Linear Regression
NOM	National Office of Meteorology
NWP	Numerical Weather Prediction
PD	Power density (kW/m^2)
PDFs	Power Density Functions
Pr	Rated Power (kW)
PVC	Present value of Cost ($)
r	Interest rate (%)
Rat_pow	Wind turbine rated power (kW)
Rat_s	Shuttle Radar Topography Mission
Rot_diam	Rated speed (m/sec)
SRTM	Rotor diameter (m)
t	Life time (year)
V	Wind speed (m/sec)
Vm	Mean wind speed (m/sec)
WAsP	Wind Atlas Analysis and Application Program

Greek letters

ηel	Electrolyzer efficiency (%)
Γ	Gamma function
ρ	Air density (kg/m^3)
σ	standard deviation (m/sec)

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ORCID

Fares Meziane http://orcid.org/0000-0003-1205-6610
Kamal Mohammedi http://orcid.org/0000-0003-3481-3627
Ilyes Nouicer http://orcid.org/0000-0001-8410-594X
Nourdine Kabouche http://orcid.org/0000-0002-1269-6303

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